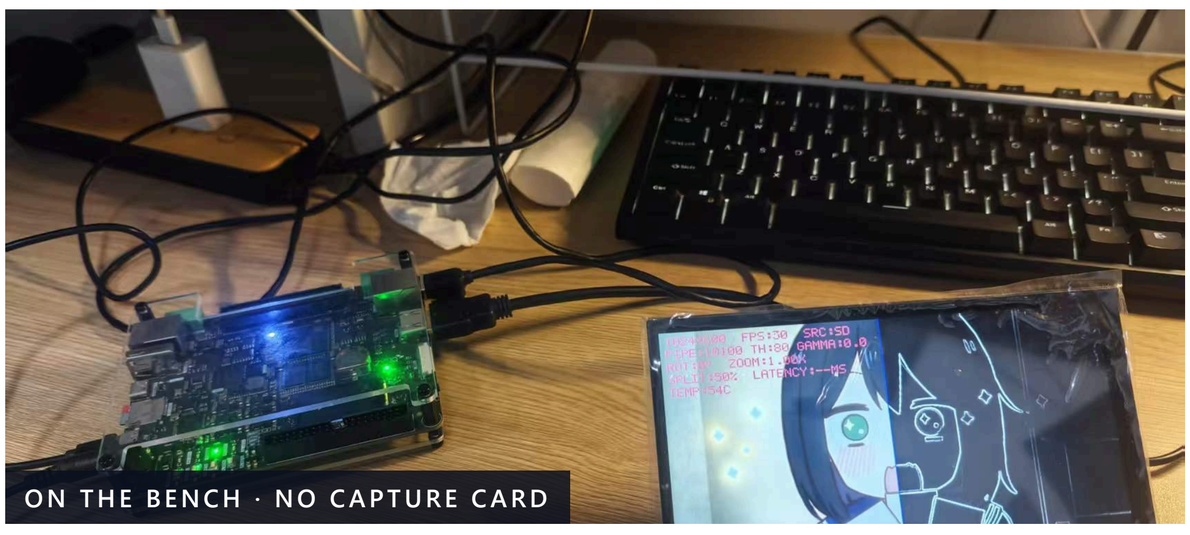
AMD

ONE FRAME. TWO PICTURES.

One read address, one clock cycle, one scanline — **the panel itself becomes the measuring instrument.** On the Ethernet path, no pixel is ever touched by software.

Device **xc7z020clg484-2** Tools **Vivado / Vitis 2025.2.1** RTL **80 files · 11,768 lines** Firmware **2,544 lines bare-metal C** On panel **r132 · bit md5 cb907d3e4898**



ON THE BENCH · NO CAPTURE CARD



THE SEAM · 50 %

Left of the seam the panel shows the **unprocessed** frame; right of it the **processed** one — same frame, same instant, one read address. Seam position is a runtime parameter (`split 0…100`), so a geometry error shows up as a step or a black wedge **on the panel**, not as a log line. Five OSD lines read `SRC · FPS · SPLIT · ANGLE · TEMP` , and the same registers are read back over serial and JTAG — three windows on one set of numbers.

165
pixel criteria · 0 FAIL
one top bench walks the whole design, per column, per clock

29.99
fps · 0 dropped
300 s / 9,000 frames, counted in hardware on the card

116.7
fps headroom
unthrottled 120 fps ingress, 960 frames, no lost byte — a bound, not a maximum

+0.687
ns WNS · 0 failing
post-route, 0 of 51,027 setup and hold endpoints failing, TNS 0

68.6
% of BRAM tiles
96 / 140 tiles · LUT 26.3 % · registers 7.6 % · DSP 8.6 %

01 WHO ASKS FOR THIS

Camera bring-up on a line
Industrial cameras push raw frames. The usual fault is a dropped packet eating part of the image, and the field only saw "it froze". Here the **drop and corruption counters live in the fabric**, read at the same instant as the picture.

Tuning an image pipeline
Comparing before/after on one frame usually costs a laptop, a capture card and a second screen — and the PC preview is not the RTL. Here it is **one seam and one control word**.

Teaching and lab work
83 benches, one command, one bitstream with an identity: the whole design can be re-run and refuted by somebody else, not only believed.

02 WHERE A PIXEL TRAVELS

Ingress
RGMII → GMII in the PL. IDDR + IDELAY, frame reassembly, port filter; ARP and ICMP answered in RTL.

Storage
DDR ping-pong. Three banks, BRAM row rings, a single-owner arbiter across ETH / SD / test card.

Geometry
Zoom · rotate · bilinear. Eight fixed-point ratios 0.25–2.00× plus auto-fit; 0–359° stays inside the panel.

Effects
Six stages, nine switches. One 9-bit control word, fixed **15-cycle** latency for any combination.

Display
A/B seam → OSD → TMDS. 1024×600 @ 59.5 Hz off a 50 MHz pixel clock, ×2 serialised.

03 WHY THE PICTURE CAN BE TRUSTED

INVARIANT

One coordinate operator
`zfit` solves the fit factor in 3 DSP48 blocks; `zmap` maps every pixel through 8, rotation running **inverse-mapped** with two linear interpolations. Eight ratios plus auto-fit, so a rotated frame **always stays inside the panel** — checked by a two-sided bench, not by an eye.

INVARIANT

Latency is a constant
Six stages, each window stage keeping its own column tag, so the chain is **15 cycles and −4 rows for every combination**. The fabric exposes one number, `u_pipe.LATENCY` ; the read scheduler never re-derives it. Across the frame boundary the read-side wrap is keyed to the ring's write beat, so the previous frame's pixels are fixed, not stale.

DISCIPLINE

Every claim can go red
A criterion with no way to fail is not evidence. **83 benches** carry a mutation control — flip one bit, delete one line, the bench must go red — and a **24-item gate** runs before release. A zero-hit scan is reported as a blind ruler, never as a pass.

SLOT · hardware vs software model
Put the panel photo taken at the same settings as figures 9–11 of the report (seam 50 %, all six effects on, rotate 45°) beside the PC reference image.

SLOT · cold boot, still owed
Switch set to QSPI, power off 10 s, power on with no PC attached: panel lighting up by itself plus the serial banner in the same frame. Until this is photographed and filed, the poster keeps saying "written and verified", not "self-start proven".

24-item gate, 0 red
one command; its last line is the verdict. Two runs inside this round went red on stale credentials — fixed in the checkers, never by loosening a criterion.

Six timing cuts declined
checksum per beat, constant hoisting, replication of the 239-fanout net, a Pblock over the FIFO BRAMs, an RGMII input window, BRAM-for-setup. Published with readings, including the ones that won one cell and lost four.

Measured, estimated, or not claimed
Junction temperature on the running board is XADC 61.2–61.7 °C (r113 board); 52.6 °C for r132 is a tool estimate at confidence Low. The QSPI image is written and byte-verified; no cold-boot capture is on file, so no auto-start claim.

```
vivado -mode batch -source build/tcl/build_system_axigpio.tcl · project → synth → impl → bitstream, 19 min 30 s measured
bash build/gates.sh · 24 pre-release items; the last line is the verdict
bash build/sim/run_one.sh tb_v98_top_seam · 165 per-pixel criteria, elapsed 03:21:55
bash build/board_verify.sh --geom --battery · geometry check + 105 serial commands against the panel
```

All readings come from one build: **bit md5 cb907d3e4898**, timing / utilisation / power reports archived with the submission.